

Oscar Bismark Tetteh

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PROFILE

I am a data scientist, statistician, and quantitative economist with over five years of experience across academia, the private sector, and the public sector. I specialize in forecasting, statistical analysis, data modelling, and applied research, with strong experience using Python, R, SQL, and data pipelines to turn complex data into clear, decision-ready insights. My work is grounded in analytical rigour, attention to detail, and a practical approach to solving problems through evidence-based analysis.

PROFESSIONAL SKILLS

- Built SQL pipelines to query, retrieve, join, and shape large datasets for analysis.
- Built ETL and ELT pipelines in Python to support data extraction, transformation, and preparation for reporting, analysis, and machine learning workflows.
- Used NumPy and pandas in Python to manipulate, clean, transform, and prepare data for analysis.
- Performed exploratory data analysis using Matplotlib and Seaborn to examine distributions, relationships, anomalies, and data structure before modelling.
- Performed qualitative and quantitative analysis using R and Python.
- Conducted predictive analysis, forecasting, statistical analysis, and hypothesis testing to support research and decision-making.
- Analysed socio-economic and financial data to identify patterns, develop forecast models, and support evidence-based decisions.
- Conducted socio-economic, regulatory, and policy impact assessments using qualitative and quantitative methods.
- Performed spatial analysis using R and QGIS.
- Evaluated qualitative and quantitative information to develop recommendations, policy options, indicators, and trend analyses.
- Created data visualizations and reporting outputs using Power BI, Excel, and Visio.
- Produced reports, presentations, and visual summaries to communicate findings clearly to technical and non-technical stakeholders.
- Configured and managed Python virtual environments using pyenv, venv, and conda in Visual Studio Code.
- Used Git and GitHub for version control, branch management, and project organization.
- Managed cross-functional projects and adapted effectively to changing priorities in fast-paced work environments.

EDUCATION

Master's in Applied Economics (August 2024)

University of Saskatchewan, Canada *Relevant Courses:* Econometrics, International Trade, Development Economics, Advanced Quantitative Methods, Microeconomic & Macroeconomic Theory, Time Series Analysis.

Certificate in Circular Economy & Implementation Science (May 2022)

University of Guelph, Canada

B.A. Economics, Statistics and Mathematics (June 2020)

University of Ghana, Ghana *Relevant Courses:* Data Analysis, Actuarial Statistics, Statistical Inference, Probability Distributions, Non-Parametric Statistics, Cost Benefit Analysis, Analysis of Experiments, Micro and Macroeconomics, Multivariate and Bi-variate Distributions.

WORK EXPERIENCE

State of Connecticut, Hartford, Connecticut

January 2024 – Present, Full-time

Associate Analyst – Department of Children and Family – Bureau of Continuous Improvement

- Re-evaluated the Structured Decision Making (SDM) model in child welfare to measure predictive performance using actuarial risk modelling, survival analysis, and applied statistical methods.
- Built SQL- and Python-based workflows to extract, join, clean, and prepare administrative and case-level data for model validation and performance assessment.
- Used pandas and NumPy to transform large datasets and support risk modelling, subgroup analysis, and research reporting.
- Conducted exploratory data analysis, statistical testing, and predictor assessment to evaluate how well the model identified future high-risk outcomes across cases and population subgroups.
- Communicated technical findings through reports, summaries, and presentations to support evidence-based improvements in child welfare decision-support tools.

Government of Alberta, Edmonton, Canada

September 2024 – December 2025, Full-time

Senior Policy and Data Analyst – Ministry of Education – Office of Statistical Information (OSI) Unit

- Developed statistical models, scenario analyses, and sensitivity analyses to forecast funding requirements and assess the impact of policy changes on budget allocations and educational equity.
- Analyzed large policy and financial datasets to identify trends, quantify potential outcomes, and support evidence-based decision-making.
- Researched and evaluated funding policy approaches across Canadian and international jurisdictions to inform improvements to Alberta's funding framework.
- Produced analytical summaries, reports, and briefing materials to communicate technical findings and recommendations to decision-makers.
- Worked independently and collaboratively on analytical, modelling, and data management activities in a fast-paced policy environment.

Government of Alberta, Edmonton, Canada

May 2024 – September 2024, Full-time

Policy and Data Analyst – Service Alberta and Red Tape Reduction – Economic & Data Unit

- Conducted forecasting and quantitative analysis on Alberta iGaming and liquor markup trends to support policy and planning decisions.
- Integrated and analyzed data from multiple sources to support dashboard development, forecasting, and market analysis.
- Tracked retail price movements across Alberta and comparator provinces and used the results to inform economic analysis and reporting.
- Supported data collection and analysis activities, including survey design and jurisdictional scans, to improve data quality and analytical usefulness.
- Prepared policy briefs, technical summaries, and presentations to communicate complex findings to diverse audiences.

University of Saskatchewan, Canada

September 2022 – April 2024, Full-time

Research & Data Analyst – Department of Economics

- Collected, extracted, cleaned, and analyzed data from Statistics Canada, the U.S. Bureau of Labor Statistics, and other databases using quantitative research methods.
- Worked with multi-source datasets to support labour market and energy price research.
- Conducted statistical analysis and interpreted findings to support research conclusions and evidence-based recommendations.
- Produced research outputs and summaries for academic and policy audiences.
- Supported applied research projects requiring data preparation, documentation, and clear communication of technical findings.

Institute of Statistical & Social Economic Research

May 2021 – May 2022, Full-time

Economic Policy and Data Analyst – Economic Policy Department

- Conducted in-depth analysis of economic trends, policy issues, and development challenges, with a focus on agriculture, rural transformation, and economic sustainability.
- Provided strategic insights and recommendations to policymakers and stakeholders based on research findings and data analysis.
- Utilized quantitative and qualitative research methods to analyse economic data, model policy scenarios, and assess the impact of policy interventions.
- Prepared policy briefs, reports, and presentations to communicate research findings, policy recommendations, and best practices to policymakers, donors, and the broader public.

Institute of Statistical & Social Economic Research

April 2020 – April 2021, Full-time

Junior Economic and Data Policy Analyst – Economic Policy Department

- Assisted senior analysts and researchers in conducting economic policy research projects, focusing on areas such as agriculture, rural transformation, and economic sustainability.
- Supported economic policy research by cleaning, organizing, and analyzing quantitative datasets using statistical methods.
- Assisted with econometric analysis, regression modelling, and policy evaluation to examine economic patterns and outcomes.
- Contributed to the development of evidence-based policy recommendations through structured data analysis and literature review.
- Helped prepare written outputs and research summaries for internal and external stakeholders.

- Gathered, cleaned, validated, and analyzed operational and market data related to cocoa production, sales, and distribution.
- Conducted exploratory data analysis to identify trends, patterns, and anomalies in commodity-related datasets.
- Supported the development and maintenance of databases used for storing and retrieving production and market data.
- Analyzed supply and demand patterns to support market trend monitoring and reporting.

RESEARCH EXPERIENCE AND OTHER SKILLS

Research Projects:

- Relationship between the Canada–US exchange rate and commodity prices.
- Gravity models of trade for EU panel data.
- Simulation-based inference using bootstrap and Monte Carlo techniques using the US exchange rate and commodity prices.
- *Master's Thesis:* Economic Determinants of Food Security in the Northwest Territories, Canada.